Assignment No -1

Title:- Install and explore the OpenGL

Theory:-

OpenGL (Open Graphics Library) is a cross-platform, hardware-accelerated, language-independent, industrial standard API for producing 3D (including 2D) graphics. Modern computers have dedicated GPU (Graphics Processing Unit) with its own memory to speed up graphics rendering. OpenGL is the software interface to graphics hardware. In other words, OpenGL graphic rendering commands issued by your applications could be directed to the graphic hardware and accelerated.

We use 3 sets of libraries in our OpenGL programs:

1. **Core OpenGL (GL)**: consists of hundreds of commands, which begin with a prefix "gl" (e.g., glColor, glVertex, glTranslate, glRotate). The Core OpenGL models an object via a set of geometric primitives such as point, line and polygon.
2. **OpenGL Utility Library (GLU)**: built on-top of the core OpenGL to provide important utilities (such as setting camera view and projection) and more building models (such as qradric surfaces and polygon tessellation). GLU commands start with a prefix "glu" (e.g., gluLookAt, gluPerspective).
3. **OpenGL Utilities Toolkit (GLUT)**: OpenGL is designed to be independent of the windowing system or operating system. GLUT is needed to interact with the Operating System (such as creating a window, handling key and mouse inputs); it also provides more building models (such as sphere and torus). GLUT commands start with a prefix of "glut" (e.g., glutCreatewindow, glutMouseFunc). GLUT is platform independent, which is built on top of platform-specific OpenGL extension such as GLX for X Window System, WGL for Microsoft Window, and AGL, CGL or Cocoa for Mac OS.
   Quoting from the opengl.org: "GLUT is designed for constructing small to medium sized OpenGL programs. While GLUT is well-suited to learning OpenGL and developing simple OpenGL applications, GLUT is not a full-featured toolkit so large applications requiring sophisticated user interfaces are better off using native window system toolkits. *GLUT is simple, easy, and small.*"
   Alternative of GLUT includes SDL, ...
4. **OpenGL Extension Wrangler Library (GLEW)**: "GLEW is a cross-platform open-source C/C++ extension loading library. GLEW provides efficient run-time mechanisms for determining which OpenGL extensions are supported on the target platform." Source and pre-build binary available at http://glew.sourceforge.net/. A standalone utility called "glewinfo.exe" (under the "bin" directory) can be used to produce the list of OpenGL functions supported by your graphics system.

OpenGL as State Machine
OpenGL operates as a *state machine*, and maintain a set of *state variables* (such as the foreground color, background color, and many more). In a state machine, once the value of a state variable is set, the value persists until a new value is given.
For example, we set the "clearing" (background) color to black *once* in initGL(). We use this setting to clear the window in the display() *repeatedly* (display() is called back whenever there is a window re-paint request) - the clearing color is not changed in the entire program.

Naming Convention for OpenGL Functions

An OpenGL functions:

- begins with lowercase gl (for core OpenGL), glu (for OpenGL Utility) or glut (for OpenGL Utility Toolkit).
- followed by the purpose of the function, in *camel case* (initial-capitalized), e.g., glColor to specify the drawing color, glVertex to define the position of a vertex.
- followed by specifications for the parameters, e.g., glColor3f takes three float parameters. glVectex2i takes two int parameters.
  (This is needed as C Language does not support function overloading. Different versions of the function need to be written for different parameter lists.)

The convention can be expressed as follows:

returnType **gl*Function*[234][sifd]** (*type value*, ...); // 2, 3 or 4 parameters

returnType **gl*Function*[234][sifd]v** (*type* **value*);   // an array parameter

Install OpenGL on Ubuntu
For installing OpenGL on Ubuntu, just execute the following command (like installing any other thing) in terminal :

sudo apt-get install freeglut3-dev

For working on Ubuntu operating system:

gcc filename.c -lGL -lGLU -lglut

where filename.c is the name of the file with which this program is saved.

Install OpenGL on windows in Code::Blocks
1. Download code block and install it
2. Go to the link and download zip file from the download link that appears after freeglut MinGW package with having link name as Download freeglut 3.0.0 for MinGW and extract it.

3. Open notepad with run as administrator and open file from
    1. This PC > C:(C-drive) > Program Files(x86) > CodeBlocks > share > CodeBlocks > templates, (then click to show All Files)
    2. Next, open glut.cbp and search all **glut32** and replace with **freeglut**.
    3. Then, open from This PC > C:(C-drive) > Program Files(x86) > CodeBlocks > share > CodeBlocks > templates > wizard > glut (then click to show All Files)
    4. Open wizard.script and here, also replace all **glut32** with **freeglut**
4. Then go to **freeglut** folder (where it was downloaded) and
    1. Include > GL and copy all four file from there
    2. Go to This PC > C:(C-drive) > Program Files(x86) > CodeBlocks > MinGW > include > GL and paste it.
    3. Then, from download folder freeglut > lib, copy two files and go to This PC > C:(C-drive) > Program Files(x86) > CodeBlocks > MinGW > lib and paste it.
    4. Again go to downloaded folder freeglut > bin and copy one file (freeglut.dll) from here and go to This PC > C:(C-drive) > Windows > SysWOW64 and paste this file.
5. Now open Code::Blocks.
    1. Select File > New > Project > GLUT project > Next.
    2. Give project title anything and then choose Next.
    3. For selecting GLUT's location : This PC > C:(C-drive) > Program Files(x86) > CodeBlocks > MinGW.
    4. Press OK > Next > Finish.

Conclusion:-

We learned steps to install OpenGL.